

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<reasoning process here> The patient presents with symptoms consistent with community-acquired pneumonia (CAP), including sore throat, fever, non-productive cough, and physical exam findings such as oropharyngeal erythema and moist rales. The chest X-ray shows patchy reticular opacities in the perihilar regions, which are classic for a bacterial infection, particularly one involving the lower respiratory tract.

Given the history and clinical picture, the most likely pathogen is *Streptococcus pneumoniae*, a common cause of community-acquired pneumonia. This organism is a Gram-positive bacterium with a cell wall containing peptidoglycan.

The immune response to *S. pneumoniae* is primarily mediated by the innate immune system, particularly through pattern recognition receptors such as Toll-like receptors (TLRs). However, TLR5 recognizes bacterial flagellin, not peptidoglycan. Peptidoglycan is recognized by TLR2, not TLR5, so option A is incorrect.

Pneumococcus is not primarily associated with a Th1 response; rather, it triggers a mixed Th1/Th2 response, but the dominant response in bacterial pneumonia is not specifically Th1. While Th1 responses are important in fighting intracellular pathogens, bacterial pneumonia like this one involves both humoral and cellular responses, but the question asks for the best description of the immune response elicited by the pathogen.

Option C: "Peptidoglycan is its major antigen that induces an immune response" â€” this is partially correct. Peptidoglycan is a major component of the bacterial cell wall and is recognized by immune cells (e.g., via TLR2 and NOD1). However, it is not the only antigen, and the immune response is not defined solely by peptidoglycan. The phrasing "major antigen" is vague and not the best descriptor of the immune response.

Option D: "This pathogen evades the immune response by encapsulation" â€” *S. pneumoniae* does have a polysaccharide capsule that helps it evade phagocytosis by macrophages, which is a well-known immune evasion mechanism. This is a key feature of *S. pneumoniae* and is relevant to the immune response.

But the question asks: "best describes the immune response elicited by the pathogen." The immune response is not just evasion â€” it's about what the body does in response. The immune response is initiated by recognition of bacterial components, including peptidoglycan, and includes inflammation and phagocytosis. However, the immune evasion (encapsulation) is a key feature of the pathogen's interaction with the immune system.

But again, the question is about the immune response *elicited* â€” not evasion. So while evasion is a feature, the immune response itself is not described by evasion.

Thus, the most accurate choice among the options must describe a key feature of the immune response.

Option C says peptidoglycan is the major antigen â€” this is true in terms of antigenic structure, but it's not a full description of the immune response.

In contrast, the immune response to *S. pneumoniae* is initiated by recognition of bacterial components such as peptidoglycan, which activates macrophages and dendritic cells.

However, none of the options perfectly describe the immune response. But the only one that directly identifies a key component of the pathogen's biology that drives the immune response is the presence